Curriculum Vitae

Jasper Cheng-Yen Chan

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EDUCATION

National Taiwan University (NTU) Sept. 2023 – June 2025

Master of Science in Computer-Aided Engineering (CAE), Department of Civil Engineering (CE)

- GPA: 4.01 / 4.30
- Advisor: Prof. Chuin-Shan (David) Chen
- Thesis: Lightweight Visual Odometry with Deep Features for Autonomous UAVs in GPS-Denied Environments: Real-World Flight Validation
- Related Coursework: Computer Vision, 3D Computer Vision with Deep Learning Applications, Applied Deep Learning, Data Structures and Algorithms, Operations Research

National Taiwan University (NTU) Sept. 2019 – June 2023

Bachelor of Science in Civil Engineering (CE)

- GPA: 3.84 / 4.30
- Ranking: 17 / 95
- Related Coursework: Introduction to Machine Learning and Deep Learning, Computer Programming

RESEARCH EXPERIENCE

Full-time Research Assistant and Project Manager, NTU Oct. 2025 – Present

Advisor: Prof. Chuin-Shan (David) Chen

Project: Online Visual-Topological Memory for Language-Guided Monocular MAV Navigation (Ongoing)

- Investigating online, traversability-aware visual-topological memory for language-guided monocular MAV navigation, distinguishing traversed edges from visually inferred loop-closure edges.
- Designing language-guided frontier exploration and a learned image-goal navigation policy to support multi-step navigation, revisiting, and exploration of unseen targets without prebuilt maps.

Project: Illumination-Robust Correspondence Fields for Dark 3D Gaussian Maps

- Developed a LiDAR-inertial pipeline for depth-supervised 3D Gaussian mapping under moving onboard illumination.
- Developed a geometry-supervised correspondence learning framework for illumination-robust feature matching, distilling correspondence and semantic features into separate 3D Gaussian fields.
- Improved cross-view matching across three dark scenes and evaluated offline odometry correction using real UAV flight data.

Project: Safe Robot Learning for Active 3D Reconstruction

- Contributed to a collaborative simulation study of hierarchical reinforcement learning for micro aerial vehicle (MAV) navigation and active 3D reconstruction.
- Investigated collision avoidance and visual tracking reliability using control barrier functions and SLAM-aware rewards.

Research Assistant and Project Manager, NTU Feb. 2023 – Oct. 2025

Advisor: Prof. Chuin-Shan (David) Chen

Project: Evaluating Visual Odometry with Deep Feature Detectors for Autonomous UAVs in Indoor Environments

- Developed a hybrid visual odometry framework combining deep feature detectors (SuperPoint and XFeat) with KLT tracking; the XFeat-based variant reduced mean ATE by 53.5% and translational RPE by 77.5% compared with VINS-Fusion on EuRoC.
- Led the integration of perception, onboard computing, and flight control for a custom-built autonomous UAV.
- Validated the system through indoor flight experiments, achieving a 90% figure-eight flight success rate with XFeat-based VO versus 40% with the baseline.

Research Assistant, NTU Sept. 2022 – Feb. 2023

Advisor: Prof. Jacob Je-Chian Lin

Project: Automated Construction Site Progress Analysis System with Schedules and Site Images

- Developed a construction progress tracking framework aligning project schedules with site images.
- Applied NLP preprocessing and LSTM/BERT models to classify schedule information for automated progress tracking.

Research Assistant, NTU Jan. 2022 – July 2022

Advisor: Prof. Ying-Chieh Chan

Project: Window View Evaluation with Virtual Reality for Urban Planning

- Developed interactive VR simulations of urban window views in Unity (C#) for participant studies.
- Designed controlled user experiments to assess perception and preferences across virtual window scenes.

PUBLICATIONS AND MANUSCRIPTS

Manuscripts Under Review

- **C.-Y. Chan**, C.-Y. Fan, W.-T. Chang, and C.-S. Chen (*2026)*, "Evaluating Deep Feature Detectors for Visual Odometry in Unmanned Aerial Vehicles: Accuracy–Robustness Trade-Offs under Real-World Flight," Computer Modeling in Engineering & Sciences, under review.

Manuscripts in Preparation

- **C.-Y. Chan**, W.-T. Chang, and C.-S. Chen, "Illumination-Robust Correspondence Fields for Dark 3D Gaussian Maps," manuscript in preparation.
- C.-Y. Fan, **C.-Y. Chan**, W.-T. Chang, and C.-S. Chen, "Autonomous UAV-Based Visual Inspection for Structural Health in Unknown Indoor Buildings," manuscript in preparation.
- D.-J. Hsu, **C.-Y. Chan**, W.-T. Chang, and C.-S. Chen, "Real-Time Hierarchical Visual Localization for MAVs in GPS-Denied Environments," manuscript in preparation.

Conference Presentations

- **C.-Y. Chan**, W.-T. Chang, and C.-S. Chen, "UAV Technologies for Civil Engineering Applications in GPS-Denied Environments," accepted for presentation at the KKHTCNN Symposium on Civil Engineering, National Taiwan University, Taipei, Taiwan, Nov. 2026.
- **C.-Y. Chan**, W.-T. Chang, and C.-S. Chen, "Autonomous UAVs in GPS-Denied Indoor Environments: System Integration, Control, and Applications," the 4th Annual Meeting and Conference of the Association of Computational Mechanics Taiwan (ACMT 2026), Taipei, Taiwan, Oct. 2026.
- C.-Y. Fan, **C.-Y. Chan**, W.-T. Chang, and C.-S. Chen, "Autonomous UAV-Based Visual Inspection for Structural Health in Unknown Indoor Buildings," the 4th Annual Meeting and Conference of the Association of Computational Mechanics Taiwan (ACMT 2026), Taipei, Taiwan, Oct. 2026.

- J.-Z. Lin, **C.-Y. Chan**, W.-T. Chang, and C.-S. Chen, "Dynamic-Aware 4D Mapping for Indoor UAV Inspection," the 4th Annual Meeting and Conference of the Association of Computational Mechanics Taiwan (ACMT 2026), Taipei, Taiwan, Oct. 2026.
- D.-J. Hsu, **C.-Y. Chan**, W.-T. Chang, and C.-S. Chen, "Cross-Platform Micro-UAV Self-Localization via Hierarchical Visual Localization on a Prior Map," the 4th Annual Meeting and Conference of the Association of Computational Mechanics Taiwan (ACMT 2026), Taipei, Taiwan, Oct. 2026.

WORK EXPERIENCE

Smart Ageing Tech Co., Ltd. June 2022 – Nov. 2022
AI and Data Science Intern

- Developed a dementia evaluation model based on a knowledge graph, improving diagnostic accuracy from 60% to 94%.
- Trained and optimized machine learning models using TF-IDF representation and parameter tuning to identify dementia risk factors.
- Deployed the model as a cloud-based API, integrating it into the company's smart nursing information system for real-time dementia risk alerts.

TECHNICAL SKILLS

- Programming: Python, C++, Java, SQL
- Robotics: ROS/ROS2, Visual Odometry, Visual SLAM, PX4, Gazebo, Camera Calibration
- Machine Learning / AI: TensorFlow, PyTorch, scikit-learn, Computer Vision
- Tools: Unity (AR/VR), Docker, AutoCAD, SketchUp

TEACHING EXPERIENCE

Teaching Assistant, National Taiwan University
Introduction to Machine Learning and Deep Learning (2024-2025)
Computer Programming (2024)

- Provided one-on-one support to students and assisted them with programming tasks.
- Designed and evaluated midterm and final projects, as well as graded programming assignments.

Math Tutor, High School Mathematics, Taipei, Taiwan (2019 – 2025)

LEADERSHIP

Treasurer, Civil Engineering Student Association, NTU
President, Music Composition Club, Wu-Ling Senior High School

HONORS AND AWARDS

- Honorable Mention Award, NASA International Space Apps Challenge (2024) — Seismic Detection Across the Solar System
- LegalTech Hackathon Finalist (2024)

LANGUAGES

English (fluent), Mandarin Chinese (native), Japanese (JLPT N2 – intermediate)